

Appendix B

Solea Model 3 Software Design Verification Protocol Data Sheets

(Partial Execution of 790-00134, Rev. R on TOP-00017 w/ LCD-2 laser controller)

(Partial Execution of 790-00134, Rev. R on TOP-00017 w/ PCON laser controller)

(Partial Execution of 790-00134, Rev. R on TOP-00051 w/ PCON laser controller)

(Rev. U of Test Report)

13 TEST METHODS AND DATA COLLECTION FORMS

Solea Model 3 Software Design Verification Protocol

13.1 DATASHEETS

TABLE OF CONTENTS

1	BACKGROUND.....	1
2	DESIGN INPUTS	1
3	REFERENCES.....	1
4	RESPONSIBILITIES.....	1
5	EQUIPMENT	2
6	MATERIALS.....	2
7	METHODS.....	2
8	DOCUMENTATION	3
9	UNIT(S) UNDER TEST.....	3
10	REPORT RESULTS.....	3
11	SAMPLE SIZE.....	4
12	ACCEPTANCE CRITERIA	4
13	TEST METHODS AND DATA COLLECTION FORMS.....	5
13.1	DATASHEETS	5
	Operating Environment Identification.....	8
	General Requirements: Installation - Upgrade	9
	SW-IN-1, SW-IN-1.1.....	9
	General Requirements: Installation - Upgrade Process.....	12
	SW-IN-2	12
	General Requirements: System Startup/Shutdown	14
	SW-SS-4.....	14
	Auditing / Troubleshooting: Error Classification & Error Handling.....	16
	SW-EH-1.2	16
	Auditing / Troubleshooting: System Diagnostic Testing.....	18

SW-SDT-1	18
State Requirements: System State Transitions.....	20
SW-SST-5.1.1, SW-SST-6.4 White Box.....	20
State Requirements: System Startup State Specific Behavior	22
SW-SIB-3.....	22
State Requirements: Standby State and Error State Specific Behavior	23
SW-LWB-4	23
State Requirements: Laser Armed State Specific Behavior	26
SW-LCB-1.....	26
Workflow Requirements: Initiation of Laser Cutting.....	29
SW-WLP-5, SW-WLP-24.1 Pulse Widths	29
Workflow Requirements: Patterns	31
SW-PA-1	31
13.2 Pattern Requirement Table (SW-PA-1).....	34
13.3 UUT Data Sheet.....	36
13.4 Equipment Data Table	38
13.5 Supplementary Testing Data Sheets.....	40
13.6 SIGNATURE IDENTIFICATION FORMS.....	41
14 DATA EVALUATION	42
15 APPROVALS	42
16 DEVIATION & FAILURE FORMS.....	43
REVISION HISTORY	45

OPERATING ENVIRONMENT IDENTIFICATION

Parameter	Expected Result	Actual Result / Data
Operating Environment Identification Section End		
Comments: <i>N/A</i> <i>DB 26-Feb-2024</i>		
Tester Signature:		Date:
Reviewer Signature:	<i>N/A</i> <i>DB 26-Feb-2024</i>	Date:

GENERAL REQUIREMENTS: INSTALLATION - UPGRADE

REQUIREMENTS

Req ID	General Requirements: Installation - Upgrade
SW-IN-1	The software shall support upgrades via installation.
SW-IN-1.1	When performing an upgrade installation, the software under test shall preserve changes made by users in the software that is being upgraded.
SW-IN-1.2	The software shall support upgrades via installation while the system is in error state.
SW-IN-1.3	The software under test shall support upgrades via installation to future software versions.
SW-IN-2	The software shall be upgradable by the user without the user having to interact with Customer Success.

SW-IN-1, SW-IN-1.1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-IN-1	Ensure the previous version(s) of the Solea SW application are installed on the system. Note: Ensure to backup (or rename) the original database (solea.db & backupsolea.db) somewhere, but then create a fresh database with the downgraded SW version prior to upgrading.	The previously released software, was successfully updated to the software-under-test.	The previously released software, was successfully updated to the software-under-test.	Passed	UUT 1, UUT 5, UUT 6	TOP-00017, PCON
SW-IN-1.1						
SW-IN-1.2	Login with 1-2-3-4 and make these custom changes:	The system is logged into.	The system is logged into.			
SW-IN-1.3	Create at least 2 different new galvo home coordinate pairs (by editing the database).	Two new galvo home coordinates pairs are created by editing the database.	Two new galvo home coordinates pairs are created by editing the database.			
SW-IN-2	Note the date listed in "LastSubscriptionCheck" value of the database.	The LastSubscriptionCheck value is recorded.	LastSubscriptionCheck is 1980-01T01:00:000			

	Input an active serial number, then restart the SW	The software is restarted.	The software is restarted.			
	Create a Favorite with the following settings: Standard Mode, 0.25mm, 15mL/min, 100% governor (maximum cutting speed).	The detailed Favorite is created.	The detailed Favorite is created.			
	Change the operator's PIN code: Navigate to the Settings window	The Settings window is open.	The Settings window is open.			
	Change the pin from 1-2-3-4 to 4-3-2-1.	The pin is changed.	The pin is changed.			
	Create a new user by selecting Add/Remove User and set the new users pin to 5-5-5-5.	The new user is created and the pin is changed.	The new user is created and the pin is changed.			
	Perform a software update through the Solea software application. Verify that the custom changes made were preserved after the SW update.	The custom changes made prior to updating were preserved after the software update. The software updated Subscriptions (if applicable) during the update process, verified by the "LastSubscriptionCheck" date in the database updating to the current date after the update.	The custom changes were preserved. LastSubscriptionCheck date matches today's date: 2/16/2024			
	Attain a client's database from the cloud and replace the current database (and backup database) with this client database. Record the serial number of the client database as serial1	The database is replaced with the client's and the serial number is recorded as serial1.	The database is replaced with a clients and the serial number is recorded 4.4.2 serial 1 = 0300788 4.5.1 serial 1 = 0300160 4.6.0 serial 1 = 0300287 4.6.1 serial 1 = 030072 4.6.2 serial 1 = 0300707 4.7.0 serial 1 = 0300481			
	Restart the software (version under test). Verify that there are no unexpected errors upon SW startup and all data migrates from the database properly.	There are no unexpected errors when using any of the clients' databases with the	There are no unexpected errors when using any of the clients' databases with the software			

		software under test and all data migrates properly.	under test and all data migrates properly.			
	Verify the system's serial number is the same as serial1.	The system's serial number is serial1.	The system's serial number is serial1.			
	Repeat this with a different client database specific to each previous SW version released to the field. Record the client database's version of SW that were used for this test.	There are no unexpected errors when using any of the clients' databases with the software under test and all data migrates properly.	There are no unexpected errors when using any of the clients' databases with the software under test and all data migrates properly.			
General Requirements: Installation - Upgrade Section End						
Comments: <i>N/A</i> <i>DB 26-Feb-2024</i>						
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>				Date: 02/22/2024 11:58 AM		
Reviewer Signature: <i>[Signature]</i>				Date: <i>2/27/2024</i>		

GENERAL REQUIREMENTS: INSTALLATION - UPGRADE PROCESS
REQUIREMENTS

Req ID	General Requirements: Installation - Upgrade Process
SW-IN-2	The software shall be upgradable by the user without the user having to interact with Customer Success.

SW-IN-2
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-IN-2	Navigate to C:\Program Files (x86)\Convergent Dental\SOLEA on the system.	The specified folder is navigated to.	The specified folder is navigated to.	Passed	UUT 1	TOP-00017, PCON
	Find and record the version of Convergent.App.PCON.communication.dll as version1.	The version of the file is recorded.	version1 = 1.0.0.0			
	Perform a software update. Record the version of software update as sw_v1.	The software successfully updated. The new SW version is recorded.	The software updated successfully. sw_v1 = 4.7.1.1011			
	Navigate to C:\Program Files (x86)\Convergent Dental\SOLEA on the system.	The specified folder is navigated to.	The specified folder is navigated to.			
	Find and record the version of Convergent.App.PCON.communication.dll as version2.	The version of the file is recorded.	version2 = 4.7.1.1011.			
	Verify version2 is the same as sw_v1.	The updated file has the same version as the newly installed software.	version2 = 4.7.1.1011 = sw_v1			

General Requirements: Installation - Upgrade Process Section End
Comments:

N/A
 DB 26-Feb-2024

Tester Signature: Rachel Scaer <i>RS 2/27/24</i>	Date: 02/21/2024 08:51 AM
Reviewer Signature: <i>[Signature]</i> <i>[Signature]</i>	Date: <i>2/27/2024</i>

GENERAL REQUIREMENTS: SYSTEM STARTUP/SHUTDOWN
REQUIREMENTS

Req ID	General Requirements: System Startup/Shutdown
SW-SS-4	The software shall support the ability to detect when a user pushes the system hardware Emergency Stop button.

SW-SS-4
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-SS-4	Login to the system as a User by touching 1-2-3-4.	The system proceeds to the home screen.	The system proceeds to the home screen.	Passed	UUT 1	TOP-00017, PCON
	Activate and emit the laser.	The laser is firing.	The laser is firing.			
	While laser is ON, press the emergency stop button.	Laser emission stops immediately	Laser emission stops immediately.			
	Release the foot pedal.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.			
	Deactivate the emergency stop button.	The interlock is cleared.	The interlock is cleared.			
	Press "Next" and "OK" in the dialogue box.	The error clears.	The error clears.			
	Verify the system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.			
	While the laser is ON, release a side panel switch.	Laser emission stops immediately.	Laser emission stops immediately.			
	Release the foot pedal.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.			
	Activate the released side panel switch.	The interlock is cleared.	The interlock is cleared.			
	Press "Next" and "OK" in the dialogue box.	The error clears.	The error clears.			

	Verify the system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.			
	While the laser is ON, disconnect the interlock from the CO2 Sensor low power board.	Laser emission stops immediately.	Laser emission stops immediately.			
	Release the foot pedal.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.	A Pop-up dialogue appears indicating the interlock is tripped, and prompting the user to deactivate the emergency stop button.			
	Re-connect the interlock to the CO2 Sensor low power board.	The interlock is cleared.	The interlock is cleared.			
	.Press "Next" and "OK" in the dialogue box.	The error clears.	The error clears.			
	Verify the system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.			
General Requirements: System Startup/Shutdown Section End						
Comments: See Equipment No. 1						
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>				Date: 02/21/2024 11:21 AM		
Reviewer Signature: <i>[Signature]</i>				Date: <i>2/27/2024</i>		

AUDITING / TROUBLESHOOTING: ERROR CLASSIFICATION & ERROR HANDLING
REQUIREMENTS

Req ID	Auditing / Troubleshooting: Error Classification & Error Handling
SW-EH-1.2	Present a notification to the user that include:

SW-EH-1.2
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UU T	System Configuratio n
SW-EH-1.2	Open the code repository under test in the Visual Studios debugger	The repository is open in the Visual Studios debugger.	The repository is open in the Visual Studios debugger.	Passe d	UU T 4	TOP-00017, PCON
	Navigate to and open MainWindowViewModel.cs.	MainWindowViewModel.cs is open.	MainWindowViewModel.cs is open.			
	Navigate to the public ICommand F4 line and update it to match the following text: <pre> public ICommand F4 { get { return _pF4 ?? (_pF4 = new DelegateCommand(delegate { _eventAggregator.GetEvent<ErrorEvent>().Publish (new SystemError(ErrorCode.FootPedalFaultMode)); </pre>	The F4 command is updated as detailed.	The F4 command is updated as detailed.			

<pre> })); } } </pre>					
Run the software from the debugger.	The Convergent UI launches.	The Convergent UI launches.			
Enter laser firing state.	The laser is firing.	The laser is firing.			
Press the F4 key.	The system throws error 3001.	The system throws error 3001.			
In the error box select next and then ok.	The error clears.	The error clears.			
Verify the system can enter laser firing state.	The system enters laser firing state.	The system enters laser firing state.			
Auditing / Troubleshooting: Error Classification & Error Handling Section End					
Comments: <i>N/A</i> <i>DB 26-Feb-2024</i>					
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>			Date: 02/21/2024 12:14 PM		
Reviewer Signature: <i>[Signature]</i>			Date: <i>2/27/2024</i>		

AUDITING / TROUBLESHOOTING: SYSTEM DIAGNOSTIC TESTING

REQUIREMENTS

Req ID	Auditing / Troubleshooting: System Diagnostic Testing
SW-SDT-1	The software shall provide the ability to the user to perform a foot pedal diagnostic test in which the user is prompted to press the foot pedal to its maximum position and view visual feedback from the software showing the foot pedal position in percent (0-100).

SW-SDT-1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-SDT-1	Login as a user with 1-2-3-4.	The system proceeds to the home screen.	The system proceeds to the home screen.	Passed	UUT 1	TOP-00017, PCON
	Go to the Settings screen and select 'Foot Pedal Test'.	The Foot Pedal Test opens.	The Foot Pedal Test opens.			
	Step on and off the foot-pedal with varying pressure.	Pressing the foot pedal displays a value between 0 and 100%, depending on the amount the foot pedal is pressed.	Pressing the foot pedal displays a value between 0 and 100%, depending on the amount the foot pedal is pressed.			
	Press "OK".	The Foot Pedal Test Closes	The Foot Pedal Test Closes.			
	Verify the system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.	The system can enter armed state and fire the laser.			

Auditing / Troubleshooting: System Diagnostic Testing Section End

Comments:

See Equipment No. 1

Tester Signature: Rachel Scaer

RS 2/27/24

Date: 02/21/2024 12:15 PM

Reviewer Signature:

[Signature]

Date:

2/27/2024

N/A Rs 2/27/24

STATE REQUIREMENTS: SYSTEM STATE TRANSITIONS
REQUIREMENTS

Req ID	State Requirements: System State Transitions
SW-SST-5	The software shall support the following state transitions for exiting the Laser Armed state:
SW-SST-5.1.1	If the foot pedal position was nonzero when entering Laser Armed state the software shall not allow transition to Laser Cutting state until the foot pedal position has returned to zero position followed by a nonzero foot pedal position.
SW-SST-6.4	Error Recovery, if a recoverable error is detected.

SW-SST-5.1.1, SW-SST-6.4 White Box
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-SST-5 SW-SST-5.1.1	Set the VerificationTestLogLevel to 4 or higher in the app config file.	The VerificationTestLogLevel is set and the app config file is saved and closed.	The VerificationTestLogLevel is set and the app config file is saved and closed.	Passed	UUT 1	TOP-00017, PCON
	Launch the Solea Software. Log into the system with the pin 1-2-3-4.	The Solea Software is launched and logged into.	The Solea Software is launched and logged into.			
	Ensure that the laser is in a suitable environment to fire the laser.	The system is setup to fire the laser.	The system is setup to fire the laser.			
	Enter Arming state and press the foot pedal prior to entering Laser Armed state.	When Laser Armed State was entered, the laser did not fire.	When Laser Armed State was entered, the laser did not fire.			
	Release the foot pedal while in Laser Armed state and press it again, verify that that this time the laser does fire.	When Laser Armed State was entered, the laser was fired.	When Laser Armed State was entered, the laser was fired.			
	Navigate to the logs. Verify we see the following logging for Laser firing: "ProcessFootpedalValue : 255, FireContinuous, LaserArmed" Note: the foot pedal value may be different	The detailed text is seen in the log file.	The detailed text is seen in the log file.			

SW-SST-6.4	Open the code repository under test in the Visual Studios debugger.	The repository is open in the Visual Studios debugger.	The repository is open in the Visual Studios debugger.	Passed	UUT 4	TOP-00017, PCON			
	Navigate to and open DeviceMonitoringServices.cs.	DeviceMonitoringServices.cs is open.	DeviceMonitoringServices.cs is open.						
	Insert the following text into line 828 "throw new Exception("test");".	The detailed text is added to line 828.	The detailed text is added to line 828.						
	Run the software from the debugger.	The Convergent UI launches.	The Convergent UI launches.						
	Enter laser firing state.	The laser is firing.	The laser is firing.						
	Verify the UI throws Error 1066.	The 1066 error is visible on the UI.	The 1066 error is visible on the UI.						
	In the error box select ok.	The error notification closes.	The error notification closes.						
	Verify the system cannot enter laser firing state.	The system is unable to enter laser firing state.	The system is unable to enter laser firing state.						
State Requirements: System State Transitions Section End									
Comments: <i>N/A</i> <i>DB 26-Feb-2024</i>									
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>				Date: 02/21/2024 11:03 AM					
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>				Date: 02/21/2024 11:09 AM					
Reviewer Signature: <i>AL SR</i>				Date: <i>2/27/2024</i>					

STATE REQUIREMENTS: SYSTEM STARTUP STATE SPECIFIC BEHAVIOR

REQUIREMENTS

Req ID	State Requirements: System Startup State Specific Behavior
SW-SIB-3	When in the Startup state and if PCON is used, the software shall verify that PCONs internal configuration matches the configuration expected by the software. Applies to PCON only.

SW-SIB-3

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-SIB-3	Navigate to the Device Manager. Set the Gemalto Cell Modem port to COM8.	The Gemalto Cell Modem port is set to COM8.	The Gemalto Cell Modem port is set to COM8.	Passed	UUT 1	TOP-00017, PCON
	Verify the PCON is connected to a port other than COM8.	The PCON is connected to a port other than COM8.	The PCON is connected to a port other than COM8.			
	Restart the system.	The system is restarted.	The system is restarted.			
	Verify PCON immediately connects to an unused port, and didn't attempt to connect to COM8.	The logs indicate the PCON was able to immediately connect to a port.	The logs indicate PCON immediately connected to port 14.			

State Requirements: System Startup State Specific Behavior Section End

Comments:

N/A

DB 26-Feb-2024

Tester Signature: Rachel Scaer

RS 2/27/24

Date: 02/21/2024 11:11 AM

Reviewer Signature:



Date:

2/27/2024

STATE REQUIREMENTS: STANDBY STATE AND ERROR STATE SPECIFIC BEHAVIOR
REQUIREMENTS

Req ID	State Requirements: Standby State and Error State Specific Behavior
SW-LWB-4	When in the Arming state, the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go into Armed state.

SW-LWB-4
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LWB-4	LCD and PCON: Setup an oscilloscope to: • Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board • Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board • Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto			Passed	UUT 2	TOP-00051, PCON
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home			

	0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		signals while in Laser Armed State. See Equipment No. 2			
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	N/A for PCON			
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard restart of the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from 0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home signals while in Laser Armed State. See Equipment No. 2	See Page 23	See Page 23	See Page 23
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	N/A for PCON			
State Requirements: Standby State and Error State Specific Behavior Section End						
Comments: See Equipment No. 2						

Tester Signature: Wesley Harrington 	Date: 02/19/2024 05:24 PM 
Reviewer Signature: 	Date: 2/27/2024

 WH 26 - Feb - 2024

STATE REQUIREMENTS: LASER ARMED STATE SPECIFIC BEHAVIOR
REQUIREMENTS

Req ID	State Requirements: Laser Armed State Specific Behavior
SW-LCB-1	When exiting in the Laser Cutting state (user fully removes foot from the foot pedal), the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go back to the Armed state.

SW-LCB-1
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LCB-1	LCD and PCON: Setup an oscilloscope to: • Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board • Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board • Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto			Passed	UUT 2	TOP-00051, PCON
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home			

	galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		signals while in Laser Armed State. See Equipment No. 2			
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	N/A for PCON			See Page 26
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard power restart on the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed State.	See Page 26		See Page 26
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	N/A for PCON		26	

State Requirements: Laser Armed State Specific Behavior Section End	
Comments: See Equipment No. 2	
Tester Signature: Wesley Harrington 	Date: 02/19/2024 05:24 PM 
Reviewer Signature: 	Date: 2/27/2024

 WH 26-Feb-2024

WORKFLOW REQUIREMENTS: INITIATION OF LASER CUTTING
REQUIREMENTS

Req ID	Workflow Requirements: Initiation of Laser Cutting
SW-WLP-5	When selecting a favorite, the software shall set and display the following cutting parameters associated with the favorite:
SW-WLP-24.1	After the delay the software shall for a configurable time linearly increase the cutting speed to a configurable percentage of the max cutting speed.

SW-WLP-5, SW-WLP-24.1 Pulse Widths
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-WLP-5	Open the code repository under test in the Visual Studios debugger	The repository is open in the Visual Studios debugger.	The repository is open in the Visual Studios debugger.	Passed	UUT 4	TOP-00017, PCON
	Navigate to and open DeviceMonitoringServices.cs	DeviceMonitoringServices.cs is open.	DeviceMonitoringServices.cs is open.			
	Navigate to line 473 with text "if (_favorite != null && _favorite.pattern != null)". Add a breakpoint at this line.	The breakpoint is added.	The breakpoint is added.			
	Run the software from the debugger.	The Convergent UI launches.	The Convergent UI launches.			
	Select a handpiece on the UI.	The debugger hits the breakpoint on line 473 and stops execution.	The debugger hits the breakpoint on line 473 and stops execution.			
	Using the variable box at the bottom of the debugger, change the value of the favorite variable to null.	Favorite has been set to null.	Favorite has been set to null.			
	Disable the breakpoint and continue execution.	The debugger continues execution.	The debugger continues execution.			
	Verify the UI throws a Favorite Update Error 1067.	The 1067 error is visible on the UI.	The 1067 error is visible on the UI.			
	In the error box select next and then ok.	The error clears.	The error clears.			
	Verify the system can enter laser firing state.	The system enters laser firing state.	The system enters laser firing state.			

SW-WLP-24.1	Set up an oscilloscope or a logic analyzer to measure laser pulse widths. Set the total view time to long enough to view the full laser ramp.	The oscilloscope is setup as detailed.	The oscilloscope is setup as detailed.	Passed	UUT 1	TOP-00017, PCON
	Login to the software as a user. Cycle through the states to enter laser armed state. Step on the foot pedal to 100% to enter laser cutting state. Hold the foot pedal until the laser ramp has completed and the cutting speed jumps to 100%. Release the foot pedal.	The laser ramp is completed, and the foot pedal is released.	The laser ramp is completed, and the foot pedal is released.			
	Using the oscilloscope Verify that the laser ramp can be seen.	The laser ramp can be seen on the oscilloscope.	The laser ramp can be seen on the oscilloscope.			
	Verify the pulse widths during the laser ramp are 10 microseconds or longer.	The pulse widths during the laser ramp is 10 microseconds or longer.	The pulse widths during the laser ramp are 10 microseconds or longer.			
Workflow Requirements: Initiation of Laser Cutting Section End						
Comments: See Equipment No. 1						
Tester Signature: Rachel Scaer RS 2/27/24				Date: 02/21/2024 10:59 AM		
Tester Signature: Rachel Scaer RS 2/27/24				Date: 02/22/2024 07:28 AM		
Reviewer Signature: 				Date: 2/27/2024		

WORKFLOW REQUIREMENTS: PATTERNS
REQUIREMENTS

Req ID	Workflow Requirements: Patterns
SW-PA-1	The software shall use the patterns as specified in the Solea Model 3 Parameters Calculator (721-00120).

SW-PA-1
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-PA-1	Complete the relevant data table in Section 13.2, by performing the following:		The following sections were completed	Passed	UUT 2	TOP-00051, PCON
	a) Keyboard required. Navigate to solea.db\patternMap. Note this 'patternMap' table contains links to the 'handpieces' table as well as the 'patterns' table via the 'patternId' and 'handpieceId' columns. Navigate to solea.db\patterns. Reference the pattern calculator (721-00120) and verify that all values listed within solea.db\patterns match the corresponding values found in the above referenced pattern calculator. Navigate to solea.db\handpieces. Verify that the values listed properly reflect the handpiece types associated with the patterns listed in the 'patternMap' table.	a) The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types.	The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types.			
	b) If using a TOP-00017, open the Solea configuration file and set the logging level to 'DEBUG'. If using a TOP-00051, the logging level can remain at 'WARN'. Restart the software and login. Select a laser pattern and verify in the log file that the software loads the correct pattern file to the laser controller	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds (minimum and	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds			

	for the selected pattern. Note: After verifying this, the logging level must be set back to 'WARN' if it is not already.	maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.	(minimum and maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.			
	Ensure the laser is setup in an appropriate environment to fire the laser. Configure an oscilloscope to measure the laser modulation signal. Set the cutting speed to the minimum setting (10%) and Touch the Turn Laser On button. Step on the foot pedal to enter Laser Cutting state. Using the oscilloscope, measure the pulse width of the laser modulation signal (after ramp-up is completed) and verify that the measured value matches the expected value per the pattern calculators referenced in Step a).		The measured value matches the expected value per the pattern calculators referenced in Step a).			
	Change the Cutting Speed to the maximum setting (100%) for the same pattern and fire the pattern again. Measure the pulse width and verify it matches the expected value per the pattern calculator.		The pulse width is measured and matches the expected value per the pattern calculator.			
	Repeat this for every Standard and Low Power spot size for Contra handpiece mode.		This is repeated for every standard and low power spot size for the contra handpiece.			
	Repeat for the 1.25mm Standard spot size for the Straight handpiece mode. Because the 'patternMap' verified in Step a.) above is		The steps are repeated and pass.			

	shown to link the exact same patterns for the Contra and Straight handpiece modes, there is no need to verify more than one Straight handpiece mode pattern.					
	Repeat for the 7.00mm spot size for the TD handpiece mode.		The steps are repeated and pass.			
	Repeat for the 2.00mm spot size for the Solea Protect handpiece (measuring the pulse width at the minimum and maximum Calibration governor values).		The steps are repeated and pass.			
	Also measure the software-controlled pattern "off-time" for any pattern that fires on/off cycles such as Palatal Coagulation and Demineralization Reduction and verify that it matches what is listed in the database, within +/- 10%.		The Coagulation and Demin off times match what is listed in the database.			
Workflow Requirements: Patterns Section End						
Comments: See Equipment No. 1						
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>					Date: 02/21/2024 08:34 AM	
Reviewer Signature: <i>[Signature]</i>					Date: 2/27/2024	

13.2 PATTERN REQUIREMENT TABLE (SW-PA-1)

Mode	Spot Size	TOP-00017 (LCD-2 / PCON) *First Value Is LCD-2, Second Value Is PCON				TOP-00051 (N/A when using LCD-2)			
		Measured Pulse Width (Min Cutting Speed) [μs]	Measured Pulse Width (Max Cutting Speed) [μs]	Measured Off-Time [ms]	Correct Pattern Executed (Log Files) [Pass/Fail]	Measured Pulse Width (Min Cutting Speed) [μs]	Measured Pulse Width (Max Cutting Speed) [μs]	Measured Off-Time [ms]	Correct Pattern Executed (Log Files) [Pass/Fail]
Contra Angle (Standard)	0.25mm	15, 15	80, 80	N/A	Pass	15.2	80	N/A	Pass
	0.50mm	28, 28	80, 80	N/A	Pass	28	80	N/A	Pass
	0.75mm	28, 28	80, 80	N/A	Pass	28	80	N/A	Pass
	1.00mm	15, 15.2	80, 80	N/A	Pass	17.2	80	N/A	Pass
	1.25mm	15, 15.2	80, 80	N/A	Pass	16	72	N/A	Pass
	1.50mm	15, 15.2	80, 80	N/A	Pass	16	72	N/A	Pass
Contra Angle (Low Power)	0.25mm	15, 15.2	80, 80	N/A	Pass	15.2	80	N/A	Pass
	0.50mm	11, 11.2	40, 40	N/A	Pass	13.2	40	N/A	Pass
	0.75mm	11, 11.2	40, 40	N/A	Pass	13.2	40	N/A	Pass
	1.00mm	11, 11.2	40, 40	N/A	Pass	13.2	40	N/A	Pass
	1.25mm	11, 11.2	40, 40	N/A	Pass	13.2	40	N/A	Pass
Straight (Standard)	1.25mm	15, 15.2	80, 80	N/A	Pass	16	72	N/A	Pass
Palatal Coagulation	7.00mm	14, 14	23, 23.2	992, 1030	Pass	15.2	24	1072	Pass

13.3 UUT DATA SHEET

Unit Under Test Number						
Item	UUT 1	UUT 2	UUT 3	UUT 4	UUT 5	UUT 6
System Serial Number	0300650	X-CAL-3	0300110	8CG1326SGY	0300109	0300261
Operating System	Windows 10	Windows 10	Windows 10	Windows 11 Business	Windows 10 Enterprise 2016 LTSB	Windows 10 Enterprise 2017 LTSB
Service Utility Version	1.6.0.154	1.6.0.152	1.6.0.154	N/A	N/A	1.6.0.154
Solea Application Version	4.7.1.1011	4.7.1.1011	4.7.1.1011	4.7.1.1011	4.7.1.1011	4.7.1.1011
Mist Board Firmware	3.9.0.2.1	3.9.0.2.1	3.9.0.2.1	N/A	3.9.0.2.2	3.9.0.2.1
Cool Board Firmware	3.9.0.2.1	3.9.0.2.1	3.9.0.2.1	N/A	3.9.0.2.2	3.9.0.2.1
Air Board Firmware	3.9.0.2.1	3.9.0.2.1	3.9.0.2.1	N/A	3.9.0.2.2	3.9.0.2.1
DCMB Board Firmware	3.9.0.2.1	3.9.0.2.1	3.9.0.2.1	N/A	3.9.0.2.1	3.9.0.2.1
LPWR Board Firmware	3.9.0.2.1	3.9.0.2.1	3.9.0.2.1	N/A	3.9.0.2.2	3.9.0.2.1
Laser Controller (LCD-2 or PCON)	PCON	PCON	LCD-2	N/A	PCON	PCON
Laser Controller Firmware	1.2.1.7	1.2.1.7	1.12.7.0	N/A	1.2.1.7	1.2.1.7
Laser Controller FPGA Version (If applicable)	1.0.0.3	1.0.0.3	7.1.4.9	N/A	1.0.0.3	1.0.0.3

System Configuration (TOP-00017 or TOP-00051)	TOP-00017	TOP-00051	TOP-00017	N/A	TOP-00017	TOP-00017
Other:	N/A	N/A	N/A	N/A	N/A	N/A

13.4 EQUIPMENT DATA TABLE

No.	Equipment Description	Manufacturer	Model / Part No.	CD Asset Tag / Serial No.	Calibration Due
1	Oscilloscope	Rigol	MSO1104Z	CD-0603 / S/N: DS1ZD192900671	May 2024
2	Oscilloscope	Rigol	DS1052E	CD-0186 / S/N: DS1ET161453740	November 2024
3					
4					
5					
6					
7		N/A			
8					
9					
10					
11					
12					
13					

14		N/A		
15			DB 26-Feb-2024	

13.5 SUPPLEMENTARY TESTING DATA SHEETS

Description of Terms and Test Instructions (As Previously Stated in Test Case):

TBD and can be attached to report.

STATE REQUIREMENTS: STANDBY STATE AND ERROR STATE SPECIFIC BEHAVIOR

REQUIREMENTS

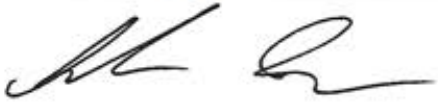
Req ID	State Requirements: Standby State and Error State Specific Behavior
SW-LWB-4	When in the Arming state, the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go into Armed state.

SW-LWB-4

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LWB-4	LCD and PCON: Setup an oscilloscope to: - Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board - Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board - Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto					
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from 0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal).	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home signals (at -0.2V) while in Laser Armed State.	Passed	UUT 3	TOP-00017, LCD-2

	Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		See Equipment No. 2			
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	The log file indicated that the software sent the galvos to the home position in Arming State, prior to entering Laser Armed State (by "SetGalvosToXyz" message in log file).			
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard restart of the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from 0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home signals while in Laser Armed State. See Equipment No. 2.			
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	The log file indicated that the software sent the galvos to the home position in Arming State, prior to entering Laser Armed State (by "SetGalvosToXyz" message in log file).			
State Requirements: Standby State and Error State Specific Behavior Section End						
Comments: See Equipment No. 2						

Tester Signature: Wesley Harrington ①	Date: 02/19/2024 04:29 PM ①
Reviewer Signature: 	Date: 2/27/2024

① WH 27-Feb-2024

STATE REQUIREMENTS: LASER ARMED STATE SPECIFIC BEHAVIOR

REQUIREMENTS

Req ID	State Requirements: Laser Armed State Specific Behavior
SW-LCB-1	When exiting in the Laser Cutting state (user fully removes foot from the foot pedal), the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go back to the Armed state.

SW-LCB-1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LCB-1	LCD and PCON: Setup an oscilloscope to: • Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board • Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board • Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home	Passed	UUT 3	TOP-00017, LCD-2

	galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		signals (at -0.2V) while in Laser Armed State. See Equipment No. 2			
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	The log file indicated that the software sent the galvos to the last calibrated Home point (by "SetGalvosToXyz" message in log file).			
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard power restart on the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal).	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home signals (at -0.2V) while in Laser Armed State.			
	Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		See Equipment No. 2			
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	The log file indicated that the software sent the galvos to the last calibrated Home point (by "SetGalvosToXyz" message in log file).			

State Requirements: Laser Armed State Specific Behavior Section End	
Comments: See Equipment No. 2	
Tester Signature: Wesley Harrington ①	Date: 02/19/2024 04:31 PM ①
Reviewer Signature: 	Date: 2/27/2024

① WH 27-Feb-2024

WORKFLOW REQUIREMENTS: PATTERNS

REQUIREMENTS

Req ID	Workflow Requirements: Patterns
SW-PA-1	The software shall use the patterns as specified in the Solea Model 3 Parameters Calculator (721-00120).

SW-PA-1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-PA-1	Complete the relevant data table in Section 13.2, by performing the following:		The following steps were performed:	Passed	UUT 3	TOP-00017, LCD-2
	a) Keyboard required. Navigate to solea.db\patternMap. Note this 'patternMap' table contains links to the 'handpieces' table as well as the 'patterns' table via the 'patternId' and 'handpieceId' columns. Navigate to solea.db\patterns. Reference the pattern calculator (721-00120) and verify that all values listed within solea.db\patterns match the corresponding values found in the above referenced pattern calculator. Navigate to solea.db\handpieces. Verify that the values listed properly reflect the handpiece types associated with the patterns listed in the 'patternMap' table.	a) The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types.	The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types.			
	b) If using a TOP-00017, open the Solea configuration file and set the logging level to 'DEBUG'. If using a TOP-00051, the logging level can remain at 'WARN'. Restart the software and login. Select a laser pattern and verify in the log file that the software loads the correct pattern file to the laser controller	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds (minimum and	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds			

	for the selected pattern. Note: After verifying this, the logging level must be set back to 'WARN' if it is not already.	maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.	(minimum and maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.			
	Ensure the laser is setup in an appropriate environment to fire the laser. Configure an oscilloscope to measure the laser modulation signal. Set the cutting speed to the minimum setting (10%) and Touch the Turn Laser On button. Step on the foot pedal to enter Laser Cutting state. Using the oscilloscope, measure the pulse width of the laser modulation signal (after ramp-up is completed) and verify that the measured value matches the expected value per the pattern calculators referenced in Step a).		The measured value matches the expected value per the pattern calculators referenced in Step a).			
	Change the Cutting Speed to the maximum setting (100%) for the same pattern and fire the pattern again. Measure the pulse width and verify it matches the expected value per the pattern calculator.		The pulse width is measured and matches the expected value per the pattern calculator.			
	Repeat this for every Standard and Low Power spot size for Contra handpiece mode.		This is repeated for every standard and low power spot size for the contra handpiece.			
	Repeat for the 1.25mm Standard spot size for the Straight handpiece mode. Because the 'patternMap' verified in Step a.) above is		The steps are repeated and pass.			

	shown to link the exact same patterns for the Contra and Straight handpiece modes, there is no need to verify more than one Straight handpiece mode pattern.					
	Repeat for the 7.00mm spot size for the TD handpiece mode.		The steps are repeated and pass.			
	Repeat for the 2.00mm spot size for the Solea Protect handpiece (measuring the pulse width at the minimum and maximum Calibration governor values).		The steps are repeated and pass.			
	Also measure the software-controlled pattern "off-time" for any pattern that fires on/off cycles such as Palatal Coagulation and Demineralization Reduction and verify that it matches what is listed in the database, within +/- 10%.		The Coagulation and Demin off times match what is listed in the database.			
Workflow Requirements: Patterns Section End						
Comments: See Equipment No. 1 See Equipment No. 2						
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>					Date: 02/21/2024 08:35 AM	
Reviewer Signature: <i>[Signature]</i>					Date: <i>2/27/2024</i>	

STATE REQUIREMENTS: STANDBY STATE AND ERROR STATE SPECIFIC BEHAVIOR
REQUIREMENTS

Req ID	State Requirements: Standby State and Error State Specific Behavior
SW-LWB-4	When in the Arming state, the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go into Armed state.

SW-LWB-4
TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LWB-4	LCD and PCON: Setup an oscilloscope to: • Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board • Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board • Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto			Passed	UUT 1	TOP-00017, PCON
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home			

	0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		positions while in Laser Armed State. See Equipment No. 2			
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	N/A for PCON			
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard restart of the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Verify on the oscilloscope that upon entering Laser Armed state the galvos were sent to their home position by verifying that the X galvo position signal increases from 0V to 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position decreases from 0V to -0.2V state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/- 0.4V) at all during this test.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed state. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent to their home positions, verified by seeing the voltage change appropriately by 0.2V upon entering Laser Armed State. The signals did not change from their galvo home signals while in Laser Armed State. See Equipment No. 2			
	LCD Only: Verify in the 'LanmarkLogging' log file that the software commanded the galvos to their home positions while in the Arming state.	LCD Only: The log file indicates that the software sends the galvos to the home position (SetGalvosToXyz) in Arming State (prior to entering Laser Armed state).	N/A for PCON			
State Requirements: Standby State and Error State Specific Behavior Section End						
Comments: See Equipment No. 2						

Tester Signature: Wesley Harrington 	Date: 02/20/2024 03:48 PM 
Reviewer Signature: 	Date: 2/27/2024

 WH 27-Feb-2027

STATE REQUIREMENTS: LASER ARMED STATE SPECIFIC BEHAVIOR

REQUIREMENTS

Req ID	State Requirements: Laser Armed State Specific Behavior
SW-LCB-1	When exiting in the Laser Cutting state (user fully removes foot from the foot pedal), the software shall set the laser "home" position to the saved x, y positions from the most recent galvo alignment operation. If no galvo alignment has been performed, the home position shall be set to 0, 0. Note: The LCD is explicitly commanded to move the galvos to the home position while the PCON is homing the galvos when commanded to go back to the Armed state.

SW-LCB-1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-LCB-1	LCD and PCON: Setup an oscilloscope to: • Probe the X galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the X galvo board • Probe the Y galvo position command signal by probing the blue (positive) side of 2-pin galvo-to-laser-controller connector on the Y galvo board • Show 200ms/div (horizontal), 0.2V/div (vertical), and trigger set to auto			Passed	UUT 1	TOP-00017, PCON
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Restart the SW. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home			

	galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.		signals while in Laser Armed State. See Equipment No. 2			
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	N/A for PCON			
	Set the Galvo X home position to +1300 and Y to -1300 within the database. Perform a hard power restart on the system. Login as a User. Touch the Turn Laser On button to enter the Arming state. Step on the foot pedal to enter Laser Cutting state and then step off it to exit Laser Cutting state. Verify on the oscilloscope that upon exiting Laser Cutting state the galvos were sent back to their home position by verifying that the X galvo position signal stays at 0.2V (note: the X galvo voltage is inverted on LCD Only) and the Y galvo position stays at -0.2V before and after the Laser Cutting state (note: if there is a lot of noise, measure from the center of the signal). Ensure that it does not go back to 0V or to double the voltage (+/-0.4V) at all during this test.	The galvos were sent back to their home positions when exiting Laser Cutting state, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed state.	The galvos were sent back to their home positions when exiting Laser Cutting State, verified on the oscilloscope. The signals did not change from their galvo home signals while in Laser Armed State.			
			See Equipment No. 2			
	LCD Only: Open the 'LanmarkLogging' log file and verify that the command "SetGalvosToXyz" has been sent while exiting Laser Cutting state.	LCD Only: The log file indicates that the SW sends the galvos to the last calibrated Home point if there is one, or to 0,0 if there is not one.	N/A for PCON			

State Requirements: Laser Armed State Specific Behavior Section End

Comments:

See Equipment No. 2

Tester Signature: Wesley Harrington ①

Date: 02/20/2024 03:47 PM ①

Reviewer Signature:



Date:

2/27/2024

① WH 27-Feb-2024

WORKFLOW REQUIREMENTS: PATTERNS

REQUIREMENTS

Req ID	Workflow Requirements: Patterns
SW-PA-1	The software shall use the patterns as specified in the Solea Model 3 Parameters Calculator (721-00120).

SW-PA-1

TESTING METHODS

Req(s)	Instruction	Expected Result	Actual Result / Data	P/F	UUT	System Configuration
SW-PA-1	Complete the relevant data table in Section 13.2, by performing the following:		The following steps were performed	Passed	UUT 1	TOP-00017, PCON
	a) Keyboard required. Navigate to solea.db\patternMap. Note this 'patternMap' table contains links to the 'handpieces' table as well as the 'patterns' table via the 'patternId' and 'handpieceId' columns. Navigate to solea.db\patterns. Reference the pattern calculator (721-00120) and verify that all values listed within solea.db\patterns match the corresponding values found in the above referenced pattern calculator. Navigate to solea.db\handpieces. Verify that the values listed properly reflect the handpiece types associated with the patterns listed in the 'patternMap' table.	a) The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types.	The 'patterns' table entries have the correct parameters as specified by the corresponding Pattern Calculator. The 'patternMap' table links the proper patterns with handpiece types. With the exception of the TOP-17 PCON Ultraguide See Deviation #1			
	b) If using a TOP-00017, open the Solea configuration file and set the logging level to 'DEBUG'. If using a TOP-00051, the logging level can remain at 'WARN'. Restart the software and login. Select a laser pattern and verify in the log file that the software loads the correct pattern file to the laser controller	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds (minimum and	b) There is a log file entry indicating the software loaded the correct pattern for each spot size selected. The measured pulse width for the minimum and maximum Cutting Speeds			

	for the selected pattern. Note: After verifying this, the logging level must be set back to 'WARN' if it is not already.	maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.	(minimum and maximum Calibration governor settings for the Solea Protect Feature) for each pattern fired match the expected values per the appropriate Pattern Calculator. The measured off-time for cycle patterns match the expected values per the appropriate Pattern Calculator.			
	Ensure the laser is setup in an appropriate environment to fire the laser. Configure an oscilloscope to measure the laser modulation signal. Set the cutting speed to the minimum setting (10%) and Touch the Turn Laser On button. Step on the foot pedal to enter Laser Cutting state. Using the oscilloscope, measure the pulse width of the laser modulation signal (after ramp-up is completed) and verify that the measured value matches the expected value per the pattern calculators referenced in Step a).		The measured value matches the expected value per the pattern calculators referenced in Step a).			
	Change the Cutting Speed to the maximum setting (100%) for the same pattern and fire the pattern again. Measure the pulse width and verify it matches the expected value per the pattern calculator.		The pulse width is measured and matches the expected value per the pattern calculator.			
	Repeat this for every Standard and Low Power spot size for Contra handpiece mode.		This is repeated for every standard and low power spot size for the contra handpiece.			
	Repeat for the 1.25mm Standard spot size for the Straight handpiece mode. Because the 'patternMap' verified in Step a.) above is		The steps are repeated and pass.			

	shown to link the exact same patterns for the Contra and Straight handpiece modes, there is no need to verify more than one Straight handpiece mode pattern.					
	Repeat for the 7.00mm spot size for the TD handpiece mode.		The steps are repeated and pass.			
	Repeat for the 2.00mm spot size for the Solea Protect handpiece (measuring the pulse width at the minimum and maximum Calibration governor values).		The steps are repeated and pass.			
	Also measure the software-controlled pattern "off-time" for any pattern that fires on/off cycles such as Palatal Coagulation and Demineralization Reduction and verify that it matches what is listed in the database, within +/- 10%.		The Coagulation and Demin off times match what is listed in the database.			
Workflow Requirements: Patterns Section End						
Comments: See Equipment No. 1						
Tester Signature: Rachel Scaer <i>RS 2/27/24</i>					Date: 02/21/2024 08:30 AM	
Reviewer Signature: <i>[Signature]</i>					Date: 2/27/2024	

=

13.6 SIGNATURE IDENTIFICATION FORMS

All personnel assigned to execute or review this verification protocol, attached deviations, modifications and/or supporting documentation, must print, sign, and initial in the space provided below. In the affiliation and the title column, fill in company name and designated title.

Printed Name	Signature	Initials	Affiliation and Title
Rachel Scaer		RS	Software Quality Engineer
Wesley Harrington		WH	Test Engineer
Dylan Burke		DB	Test Engineer
Andreas Johansson		AJ	Dir. Software Eng.

14 DATA EVALUATION

1. Data is evaluated on a pass/fail basis regarding the acceptance criteria of the requirements under test. The results from this protocol can be analyzed by evaluating the results of the completed datasheets following a testing execution.

15 APPROVALS

1. Signatures on file electronically.
2. Engineers responsible for executing the protocol must read and understand the protocol prior to execution.

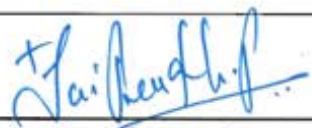
16 DEVIATION & FAILURE FORMS

<u>Deviation Form</u>	
Deviation Number:	
Date of Deviation	Document Number Section/Step # Applicable to Deviation
Plan Requirement:	
Description of Deviation:	
Comments/Rationale:	
Completed By: _____ Date: _____ (Signature)	
Technical Expert (or designee):	
Approved By: _____ Date: _____ (Signature)	
Quality Assurance:	
Approved By: _____ Date: _____ (Signature)	

<u>Failure/Out of Specification Form</u>	
Failure Number:	
Date of Failure	Document Number Section/Step # Applicable to Failure/Specification
Plan Requirement:	
Description of Failure:	
Comments/Root Cause:	
Response/Corrective Action:	
Completed By: _____ Date: _____ (Signature)	
Technical Expert (or designee): Approved By: _____ Date: _____ (Signature)	
Quality Assurance: Approved By: _____ Date: _____ (Signature)	

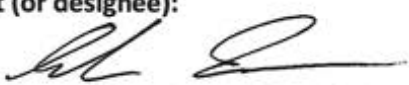
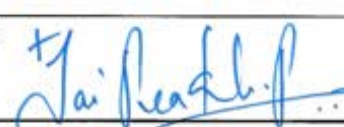
Deviation Form

Deviation Number:01

Date of Deviation	Document Number Section/Step # Applicable to Deviation
2/22/24	Solea Model 3 Software Design Verification Protocol 790-00134 Rev. R
Plan Requirement:	
SW-PA-1	
Description of Deviation:	
The Ultra Guide Pattern: CUS-01741_PCON_TOP-00017_Hp025mmRevA.coe has 80 points in 721-00120_Rev_L_-_Solea_Model_3_Pattern_Parameter_Calculator but has 1 point in the 4.7.1 database.	
Comments/Rationale:	
Ultraguide is not currently implemented on Model 3, meaning the Pattern Calculator and database discrepancy doesn't affect the test case. The Pattern Calculator is being updated in 721-00120_Rev_M_-_Solea_Model_3_Pattern_Parameter_Calculator to reflect 1 point in the Ultraguide pattern, matching the database.	
Completed By: <u></u> Date: <u>2/27/24</u> (Signature)	
Technical Expert (or designee): Approved By: <u></u> Date: <u>2/27/2024</u> (Signature)	
Quality Assurance: Approved By: <u></u> Date: <u>27 FEB 2024</u> (Signature)	

Deviation Form

Deviation Number: 02

Date of Deviation	Document Number Section/Step # Applicable to Deviation
2/22/24	Solea Model 3 Software Design Verification Protocol 790-00134 Rev. R
Plan Requirement:	
SW-IN-1	
Description of Deviation:	
Future updates were performed on a system running 4.7.1. One update was in error state, and the other was not. All custom changes were preserved after each update.	
Comments/Rationale:	
These steps are in the previous versions of SW-IN-1 test case but we're accidentally removed from the most recent version of the test case. These steps are being restored in SW-IN-1 test case in the next version of Solea Model 3 Software Design Verification Protocol 790-00134 Rev. T. Requirements, SW-IN-1.2 and SW-IN-1.3 are also being added to 822-00007 Rev. W Solea Model 3 Software Requirements Document. These requirements specify that future updates shall be able to be performed in and out of error state.	
Completed By:  Date: 2/27/24 (Signature)	
Technical Expert (or designee): Approved By:  Date: 2/27/2024 2/27/2024 D.2 (Signature)	
Quality Assurance:  Date: 27 FEB 2024 (Signature)	

REVISION HISTORY

Revision Level	Revision Date	DCO/ECO Number	Description of Revision	Revision Author
A	19-Mar-2018	DCO-00913	Initial Release	O. Hradyska
B	10-Apr-2018	DCO-00924	Edited test cases for accuracy, and per revision to the software requirements. Minor corrections throughout the protocol were made.	O. Hradyska
C	23-Aug-2018	DCO-00994	Formatting Fixes.	C. Woodmansee
D	18-Oct-2018	DCO-01050	Edited test cases for accuracy and per revision update to Solea Model 3.0 Software Requirements, 822-00007 Rev C.	C. Woodmansee
E	08-Apr-2019	DCO-01146	Edited test cases for accuracy per modification forms from previous execution.	C. Woodmansee
F	30-May-2019	DCO-01169	Edited test cases for accuracy per revision update to Solea Model 3.0 Software Requirements, 822-00007 Rev D.	C. Smaldone
G	27-Nov-2019	DCO-01308	Edited test cases for accuracy per modification and deviation forms from previous execution.	N. Sylvia
H	20-Feb-2020	DCO-01367	Merge tests relating to debounce of pneumatics errors in Appendix 1.	N. Sylvia
I	07-Jul-2020	DCO-01434	- Modified protocol layout to match the Verification Test Protocol Template (731-00006 Rev. B). - Updated for accuracy per JIRA PS-1160. - Updated per the Solea Model 3 Software Requirements Document (822-00007 Rev. I).	C. Smaldone
J	02-Oct-2020	DCO-01479	- Updated for accuracy per JIRA PI-486. - Updated per the Solea Model 3 Software Requirements Document (822-00007 Rev. M) for removal of RFID feature and related requirements.	C. Smaldone
K	28-Apr-2021	DCO-01568	- Updated for accuracy per JIRA PI-491. - Updated per the Solea Model 3 Software Requirements Document	C. Smaldone

Revision Level	Revision Date	DCO/ECO Number	Description of Revision	Revision Author
			(822-00007 Rev. N) for the addition of Cavity Prevention requirements.	
L	16-Jun-2021	DCO-01585	- Updated for accuracy per JIRA PI-536. - Updated per the Solea Model 3 Software Requirements Document (822-00007 Rev. O) for the addition of System Configuration Management (Backup & Restore) and Broadcast Messaging requirements.	C. Smaldone
M	28-Oct-2021	ECO-02052	- Updated for accuracy per JIRA PI-536. - Added rationale for sample size requirement.	C. Smaldone
N	01-Feb-2022	DCO-01664	- Updated for accuracy per JIRA PI-545. - Updated per the Solea Model 3 Software Requirements Document (822-00007 Rev. P) for the addition the Demineralization Reduction (formerly Cavity Prevention). - The Sleep feature was renamed to Palatal Coagulation or variations thereof and handpiece renamed to TD to support the De Novo submission for Demineralization Reduction.	C. Smaldone
O	08-Apr-2022	DCO-01710	Updated the IFU in section 1 (Background) to include Demineralization Reduction.	C. Smaldone
P	09-Mar-2023	DCO-01827	Formatting Updates (No content changes)	C. Woodmansee
Q	Apr-2023	DCO-01837	- Updated for accuracy per JIRA PI-549 - Updated per the Solea Model 3 Software Requirements Document (822-00007 Rev. U) - Deleted SW-FS-x, SW-TL-x, SW-SLS-4, SW-SRS-6, SW-DRS-4, SW-DRS-6, SW-AR-5.15, SW-WLP-31.4 test cases - Added SW-DM-x, SW-MHR-x test cases	W. Harrington
R	Feb-2024	DCO-01943	Updated the following test cases to reflect changes in SW v4.7.1: SW-IN-1.1, SW-SST-5.1.1, SW-SS-4, SW-SDT-1, SW-LCB-1, SW-LWB-4	R. Scaer
S	Feb-2024	ECO-02270	Updated SW-IN-1 test case to reflect new requirements SW-IN-1.2 and SW-IN-1.3	R. Scaer